Stochastic Processes Semester 1

Tutorial 6 Extension Questions – Solutions

1. (a) $\mathbf{P}(X_1 - X_2 - X_3 = 0) = \sum_{i=1}^{\infty}(i-1)q^{2i-3}p^3$.
(Hint: $\mathbf{P}(X_1 - X_2 - X_3 = 0) = \mathbf{P}(X_1 = i, (X_2 + X_3) = i)$)

(b)
$$\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = 2k) = \frac{2(k-1)}{(2k-1)(2k-2)}.$$

(c)
$$\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = 2l) = \frac{2(2l-k-1)}{(2l-1)(2l-2)}.$$

(d) The value of sum X_1, X_2, X_3 **depends** on p, but the probability $\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = n)$ $(n > k)$ **does not** depend on p

2. (i) Since Y_1 and Y_2 are independent,
$$\mathbf{P}(Z \geq x) = e^{-\alpha_1 x} \times e^{-\alpha_2 x}.$$
(Hint: $\mathbf{P}(Z \geq x) = \mathbf{P}(Y_1 \geq x)\mathbf{P}(Y_2 \geq x)$)

(ii) (Hint: Note that this is the distribution of $Y_1 + Y_2$)

$$\begin{aligned}
f_T(t) &= \mathbf{P}(Y_1 + Y_2 \leq t) \\
&= \int_0^{\infty} f_{Y_1}(x) f_{Y_2}(t-x)dx \quad \text{(result seen in lectures)} \\
&= \int_0^{t} f_{Y_1}(x) f_{Y_2}(t-x)dx \quad \text{(since } f_{Y_1}(x) = 0 \text{ for } x < 0 \text{ and } f_{Y_2}(t-x) = 0 \text{ for } x > t\text{)} \\
&= \int_0^{t} \alpha_1 e^{-\alpha_1 x} \times \alpha_2 e^{-\alpha_2 (t-x)}\, dx \\
&= \frac{\alpha_1 \alpha_2}{\alpha_1 - \alpha_2} \left(e^{-\alpha_2 t} - e^{-\alpha_1 t} \right).
\end{aligned}$$